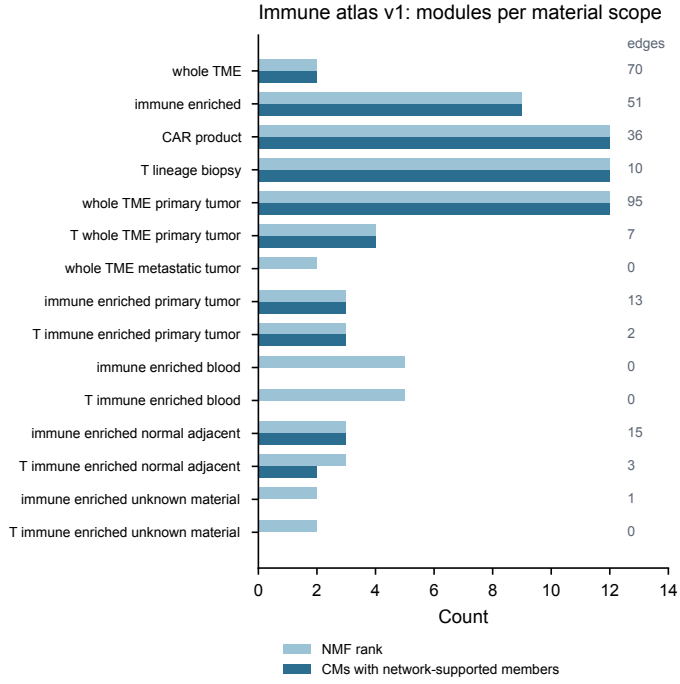
a



b

Three layers to compare

Composition CM (non-epithelial nodes only)

immune–stromal–vascular frequency; malignant programme as context

Composition CM (+ malignant / epithelial nodes)

tumour-specific nodes per cancer; CNV-qualified, split by cancer

State layer inside identity (immune datasets)

patient × identity × programme: mean, upper quantile, active fraction

Compare on common samples, same K and seeds;
report matched / split / merged modules.